
4MT-VLM: How Coarse Is a VLMs Cognitive Map?

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Abstract

1 An agent that moves must recognise a place from a viewpoint it has never seen.
2 We introduce 4MT-VLM, a dataset of procedurally generated landscapes, each
3 rendered across five stimulus modes that remove appearance cues while holding
4 layout fixed: shape and colour, shape only, colour only, bare terrain peaks with
5 no objects, and a valley viewpoint that puts the peaks on the horizon. The last
6 condition is commonly used in clinics to probe hippocampal function in human
7 patients. We test this benchmark across sixteen different open and closed-source
8 models and report 4AFC performance, a measure which is also used to grade
9 human participants. We observe that models identify a place from the studied
10 viewpoint but lose it once the camera moves, dropping below the 25% chance
11 level at 135° where a human observer scores 85%. Frontier models (Gemini 3.8
12 Flash, GPT-5.6) answer only 39% and 31% of rotated trials correctly, recovering to
13 85% and 55% only when distractors are moved more than 30 meters apart. Our
14 benchmark demonstrates that while current VLMs possess rudimentary cognitive
15 maps, their spatial resolution remains fundamentally too coarse to maintain a stable,
16 3D understanding of the world once the viewpoint changes.

17 1 Introduction

18 An agent that moves must recognise a place from a viewpoint it has never seen. Doing so requires an
19 allocentric representation, i.e. a cognitive map of where things are that does not depend on where the
20 observer stands [8, 11]. As vision-language models (VLMs) are increasingly deployed as the visual
21 reasoning engines for embodied agents, they are expected to maintain a stable, 3D understanding of
22 the world as they navigate and their camera view changes.

23 How well they actually do this is hard to read off current spatial benchmarks. Most multimodal
24 evaluations score perception, language, and geometry together, returning a single accuracy number. A
25 low score indicates that something is missing without identifying what, and a high score often means
26 the model found a visual shortcut, such as matching textures or background colours. Furthermore,
27 difficulty in these benchmarks is arbitrary. Because embodied agents operate in physical space, their
28 spatial reasoning failures should be measurable in physical units, not just percentage points on a static
29 dataset.

30 To build a more diagnostic evaluation, we adapt The Four Mountains Test (4MT) [6], which was
31 designed specifically to isolate allocentric spatial memory. A participant studies a computer-generated
32 landscape, then must identify it among four candidates rendered from a new viewpoint, with all colours
33 and textures resampled. Scores fall with hippocampal damage [1, 6] and in pre-dementia Alzheimer’s
34 disease [2, 12], which attacks the exact brain regions responsible for allocentric navigation.

35 For benchmarking allocentric perception, we build a novel test with two primary constraints:

- 36 1. **Viewpoint change** $\Delta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ\}$. Appearance is resampled at every Δ ,
37 so $\Delta = 0$ is not an image match but a check that the place can be identified at all. We call

accuracy at $\Delta = 0$ the *appearance gate*, and treat it as the denominator for any claim about viewpoint invariance.

2. **Distractor similarity, in metres.** For every pair of scenes we compute the rotation-optimal layout distance $D(i, j) = \min_{\phi} \frac{1}{K} \sum_k \|R(\phi)p_{ik} - p_{jk}\|$, which is what is left after the best rigid rotation, and draw distractors from a chosen percentile band of that distribution.

Crossing these two axes separates three distinct failures that standard benchmarks run together: failing to identify the scene visually, failing to compensate for camera rotation, and failing to distinguish physical layouts that are too close together. Our contributions are:

- **4MT-VLM:** A diagnostic benchmark for embodied AI, adapted from a clinical test of hippocampal function. It contains 500 trials over 100 procedurally generated landscapes, systematically stripping away appearance cues to isolate layout across graded viewpoint shifts.
- **Evaluation in physical units.** We generate metric “hard negatives” by drawing distractors from percentile bands of layout distance in metres. Because we redraw only the distractors while leaving the target and azimuths identical, differences in performance represent a measurable limit on the model’s spatial resolution.
- **A physical diagnosis of scaling failure.** We show that scaling open-weight models from 1B to 235B parameters drastically improves scene identification (the appearance gate) but does nothing for rotational invariance, leaving performance at or below chance. Distance is the only manipulation that recovers performance: Gemini 3.8 Flash jumps from 39% to 85% accuracy only when distractors are moved 31 m apart. The information is there, but its resolution is fundamentally too coarse for precise allocentric navigation.

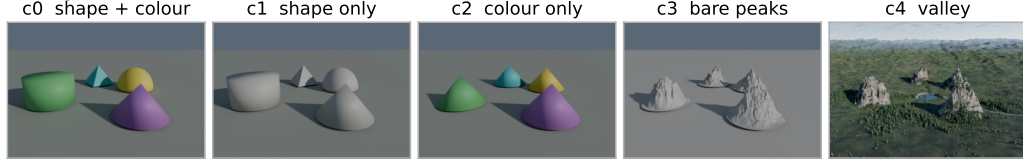
2 Related work

Spatial memory in humans. The 4MT turns the cognitive map account of hippocampal function [8] into a four-alternative forced choice, and its diagnostic value comes from the viewpoint shift specifically [6]. Separately, Shepard and Metzler [10] showed that the time to match two objects across a rotation grows linearly with angular disparity, which indicates a transformation applied to a representation rather than a lookup of a stored view. Both paradigms fix the scene, move the vantage point, and measure what that costs. Frey et al. [3] take the same task to artificial networks, training them on a scene-perception problem while reading out what the learned representations encode.

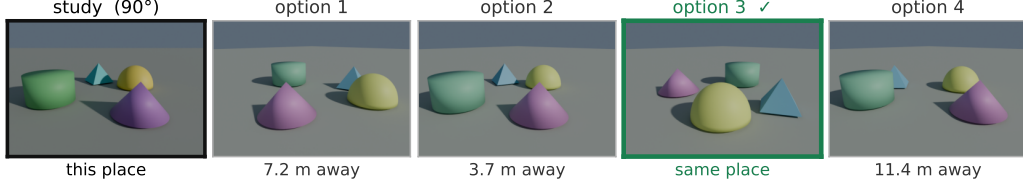
Spatial benchmarks for vision–language models. Recent evaluations keep finding the same split between egocentric and allocentric questions. Fu et al. [4] turn fourteen classic vision tasks into multiple choice and report frontier accuracy near 50% where people score above 95%. Yang et al. [13] measure spatial memory from video and find that chain-of-thought, self-consistency and tree-of-thoughts all fail to help. Li et al. [7] find competent reasoning in the camera frame and failure in another entity’s frame. Zhang et al. [15] isolate perspective taking and rotation across 43 models and report a strong egocentric bias against 91% human accuracy, and Zhang et al. [14] find that models handle 2D relations within one image but not the combination of several views into one global frame. These benchmarks establish that the capability is weak. Difficulty in them comes from the choice of question, not from a measured property of the individual item, so they can report that a model fails without locating the point where it starts to.

Building allocentric representations. A parallel line of work supplies the missing representation instead of measuring it. Gu et al. [5] build a voxelised cognitive map from video and fuse it back into the visual features. Ruan et al. [9] build a top-down landmark tree using reconstruction and segmentation tools, and report that a text-only model reading that tree comes close to multimodal performance. Both assume the representation is missing and add it from outside. We ask instead what the representation already inside the model supports, and where it runs out of resolution.

One place, five stimulus modes



One trial in c0 shape + colour



One trial in c4 valley



Figure 1: Top: one place in the five stimulus modes, at one viewpoint. Colour is removed, then shape, then the objects themselves, and c4 moves the camera into the valley so the peaks sit on the horizon. Layout is identical across the five. **Middle and bottom:** one trial in c0 and one in c4. The study view, then four candidates rendered 135° away with appearance resampled, labelled with their layout distance to the target.

3 Methods

Stimuli. We rendered 100 procedurally generated landscapes of four peaks each in Blender, from eight azimuths under two appearance samples. Five stimulus *modes* strip appearance cues away while holding layout fixed: shape and colour (c0), shape only (c1), colour only (c2), bare terrain peaks with no objects (c3), and a valley viewpoint with the peaks on the horizon (c4) (see Figure 1). All modes draw from a shared inventory of objects, so that no mode carries landmark identity that another lacks.

Task. Each trial shows a study image and four candidates rendered at a common test azimuth, one of which is the study scene. Study and test images always use different appearance samples, at every Δ . The full set is 500 trials, balanced over 5 modes $\times$ 5 values of Δ $\times$ 20 items, with the correct answer spread evenly over the four response positions.

Distractor sets. Distractors come from a percentile band of $D(i, j)$, computed separately within each mode because the modes differ in absolute scale. The main set uses the 0–10th percentile, which gives a median nearest-distractor distance of 6.8 m. Two matched sets redraw only the distractors, from the 40–60th percentile (31.4 m) and the 90–100th (42.8 m).

Landmark count. The distractor sets vary how far the alternatives sit from the target while holding the scenes fixed. A second manipulation varies the number of objects in the scenes. We rendered three further banks of 100 landscapes containing one, two and six landmarks, identical to the main bank in every render setting (eight azimuths, 640×440 , 24 samples, same seed), and built a matched four-landmark bank from the main scenes so that all four sizes are constructed the same way. These use the object mode (c0) only, because every landmark must be unique in both shape and colour and the inventory holds six of each.

Observer	Params	Accuracy	95% CI	$\Delta = 0$ gate	$\Delta \geq 45$
Human	–	86%	[78%, 91%]	100%	82%
InternVL3.5-4B	4B	17%	[11%, 26%]	35%	12%
InternVL3.5-2B	2B	21%	[14%, 30%]	25%	20%
InternVL3.5-1B	1B	23%	[16%, 32%]	35%	20%
InternVL3.5-38B	38B	28%	[20%, 37%]	65%	19%
InternVL3.5-8B	8B	28%	[20%, 37%]	55%	21%
InternVL3.5-30B-A3B	30B	29%	[21%, 39%]	55%	22%
Qwen2.5-VL-32B	32B	29%	[21%, 39%]	75%	18%
Qwen2.5-VL-3B	3B	29%	[21%, 39%]	30%	29%
Qwen3-VL-235B (t)	235B	30%	[22%, 40%]	75%	19%
InternVL3.5-14B	14B	31%	[23%, 41%]	50%	26%
Qwen2.5-VL-72B	72B	31%	[23%, 41%]	75%	20%
Qwen3-VL-235B (i)	235B	33%	[25%, 43%]	90%	19%
InternVL3-38B	38B	34%	[25%, 44%]	85%	21%
Qwen2.5-VL-7B	7B	36%	[27%, 46%]	55%	31%
Gemini 3.8 Flash	–	44%	[35%, 54%]	65%	39%
GPT-5.6 Luna	–	45%	[36%, 55%]	100%	31%

Table 1: Four-alternative forced choice with hard foils (100 trials, chance 25%: 20 unrotated and 80 rotated). The $\Delta = 0$ column is the appearance gate. Qwen3-VL-235B (i) is the *instruct* variant, (t) the *thinking* one. Rows are ordered by overall accuracy. Bold marks the human row and the best model in each column.

Procedure. Every observer answered the same 100-trial subset, stratified over mode and Δ . Models got a chain-of-thought instruction, a token budget of 8000 and, where the provider exposes it, medium reasoning effort. Every instruction is reproduced verbatim in the appendix.

Analysis. Chance level for the four-alternative forced choice (4AFC) task is 25%. Confidence intervals are Wilson intervals, paired comparisons on identical trials use the McNemar test, and distributional comparisons use permutation tests with 2×10^4 resamples. We checked that no observer’s answer distribution alone gives an advantage: the accuracy expected from each observer’s own answer frequencies, ignoring the images, is between 19% and 27% for all seventeen observers.

4 Results

We report two accuracies throughout. The appearance gate is the accuracy at $\Delta = 0$, where the place is shown from the studied viewpoint under a fresh appearance sample, e.g. the scene did not change but the position of the sun did. The rotated accuracy is the performance of the model for all other images where the viewpoint change is $\Delta \geq 45^\circ$.

Models identify the scene and lose it under rotation. Table 1 reports the two accuracies separately. GPT-5.6 Luna answers 100% of unrotated trials and 31% of rotated trials. Qwen3-VL-235B answers 90% and 19%, where a human observer answers 100% and 82%. On the identical 100 trials, paired exact McNemar puts every model below the human baseline ($p = 1.3 \times 10^{-10}$ for Gemini 3.8 Flash, 1.3×10^{-8} for GPT-5.6 Luna, 2.3×10^{-14} for Qwen3-VL-235B), and the gap holds within rotated trials alone ($p \leq 1.3 \times 10^{-8}$ for all sixteen). The task is passable, and the viewpoint change is the part the models fail.

Scale buys scene identification, not viewpoint invariance. Across Qwen2.5-VL at 3B, 7B, 32B and 72B, with prompt, stimuli and decoding fixed, unrotated accuracy rises $30 \rightarrow 55 \rightarrow 75 \rightarrow 75\%$ while rotated accuracy falls $29 \rightarrow 31 \rightarrow 18 \rightarrow 20\%$ and crosses below chance (Figure 2). To rule out a floor effect from distractor difficulty, we repeated the series with distractors 4.6 to $6.3\times$ further away. Rotated accuracy stays flat at $24 \rightarrow 27 \rightarrow 26 \rightarrow 28\%$. On the same trials the identical change takes Gemini 3.8 Flash from 39% to 85% and GPT-5.6 Luna from 31% to 55%. Another generation at three times the size does not change the picture. Qwen3-VL-235B-A22B identifies 90% of unrotated scenes and answers 19% of rotated ones; widening the distractors takes its gate to 100% and its rotated accuracy to 25%, which is chance. More inference-time computation does not help either: the *thinking* variant reaches 30% overall against 33% for *instruct*, with the same 19% rotated.

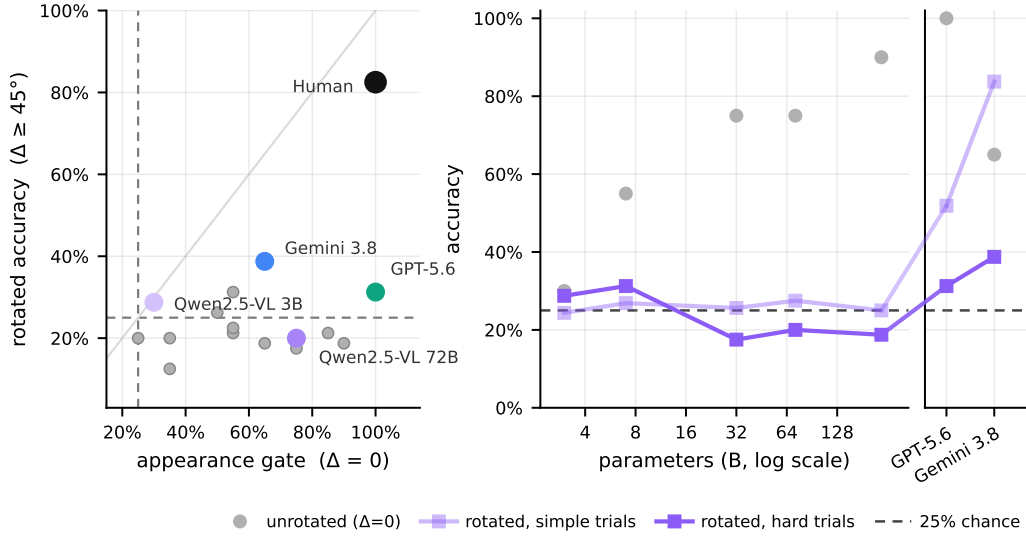


Figure 2: Left: unrotated against rotated accuracy. An observer working from appearance sits in the lower right; one that cannot identify the scene at all sits in the lower left. Grey points are the models tested but without labels in this figure. **Right:** the same two quantities against parameter count, on the identical trials, for Qwen2.5-VL at 3B, 7B, 32B and 72B and Qwen3-VL-235B in instruct mode. Grey is unrotated accuracy, which rises with scale while rotated accuracy does not, on hard trials or on simple ones. InternVL3.5 runs the same ladder and lands on top of the Qwen line and is reported in Table 1.

To test the scaling beyond the Qwen family, we ran InternVL3.5 at 1B, 2B, 4B, 8B, 14B and 38B on the same trials with the same prompt and token ceiling. Unrotated accuracy rises from 25% to 65% between 2B and 38B, while rotated accuracy reads 20, 20, 12, 21, 26, 19% across the whole $38\times$ range and never clears the 25% chance level. Its mixture-of-experts member (30B total, 3B active) behaves like its dense neighbours, at 55% and 22%. Across all fourteen open-weight models in the set, spanning 1B to 235B parameters, no rotated accuracy passes 31% (chance at 25%).

The worst viewpoint change across model is 135°. Pooled over the sixteen models, accuracy at 135° is $48/320 = 15.0\%$, clearly below chance (one-sided binomial $p = 9 \times 10^{-6}$), and recovers to 23% at 180° (Table 2). Taking each model’s own worst viewpoint, seven bottom out at 135°, three at 90°, one at 45° and one at 180°, and four tie across two viewpoints. A 180° turn is the largest change and the condition that should be hardest if accuracy simply decayed with angle. It suggests the models are applying an image heuristic, such as matching against a reflection of the original scene, which partially succeeds at a direct reversal but misleads the model at intermediate angles.

Distractor separation separates frontier from open models. Raising the median nearest-distractor distance from 6.8 m to 31.4 m on identical trials takes Gemini 3.8 Flash from 39% to 85% (exact McNemar $p = 1.5 \times 10^{-10}$) and GPT-5.6 Luna from 31% to 55% ($p = 2 \times 10^{-3}$; Figure 2). The same change moves no open model significantly at either wider separation. From 6.8 m to 31.4 m, Qwen2.5-VL at 3B, 7B, 32B and 72B moves by -4 , -6 , $+10$ and $+4$ points ($p = 0.65, 0.36, 0.13, 0.63$), and InternVL3-38B by -3 ($p = 0.66$). From 6.8 m to 42.8 m the largest shift in the open set is $+11$ points, for Qwen2.5-VL-72B ($p = 0.08$).

Making the task easier raises the gate and leaves rotation alone. Varying the number of landmarks gives a second route to the same conclusion, with different scenes rather than different distractors (Figure 3). Qwen2.5-VL-72B goes from identifying 30% of one-landmark scenes to 100% of six-landmark scenes, and its rotated accuracy over the same range moves from 12% to 19%. InternVL3.5-38B behaves the same way, 55% to 75% unrotated against 15% to 21% rotated. GPT-5.6 Luna identifies every scene at every size, while its rotated accuracy rises from 15% to 75%.

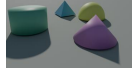
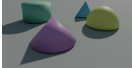
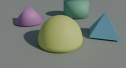
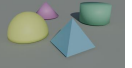
					
Observer	$\Delta = 0^\circ$	$\Delta = 45^\circ$	$\Delta = 90^\circ$	$\Delta = 135^\circ$	$\Delta = 180^\circ$
Human	100%	90%	80%	85%	75%
InternVL3.5-4B	35%	20%	20%	5%	5%
InternVL3.5-2B	25%	10%	25%	20%	25%
InternVL3.5-1B	35%	15%	30%	25%	10%
InternVL3.5-38B	65%	15%	15%	25%	20%
InternVL3.5-8B	55%	40%	20%	5%	20%
InternVL3.5-30B-A3B	55%	35%	15%	25%	15%
Qwen2.5-VL-32B	75%	30%	15%	0%	25%
Qwen2.5-VL-3B	30%	50%	15%	25%	25%
Qwen3-VL-235B (t)	75%	30%	25%	10%	10%
InternVL3.5-14B	50%	30%	25%	20%	30%
Qwen2.5-VL-72B	75%	40%	15%	10%	15%
Qwen3-VL-235B (i)	90%	40%	15%	5%	15%
InternVL3-38B	85%	40%	5%	15%	25%
Qwen2.5-VL-7B	55%	45%	15%	25%	40%
Gemini 3.8 Flash	65%	45%	50%	15%	45%
GPT-5.6 Luna	100%	35%	35%	10%	45%
All models pooled	61%	32%	21%	15%	23%

Table 2: Accuracy by viewpoint change. The images above the columns are one c0 scene at each viewpoint change, from the same study view. Pooled over the models the floor is $\Delta = 135^\circ$, below chance, and accuracy recovers at a half turn. Bold marks each model’s own worst viewpoint; rows that tie for worst are left unmarked.

No instruction style recovers the missing operation. Qwen2.5-VL-32B answered the same 80 rotated trials under six instructions, with stimuli and decoding held fixed (see Appendix A.1 for exact prompts). *cot*, used for every other run in the paper, says the matching option may be viewed from the study direction or from a different one, while *cot + view* states that it is viewed from a different viewpoint. The other four keep that framing and add an explicit strategy: estimate the rotation angle and mentally rotate the peaks, checking whether their clockwise order is preserved (*mental_rotation*); take the most distinctive peak as an origin and trace the bearings and distances of the other three from it (*anchor*); imagine the layout from directly above and work out where the camera would stand on that map (*birdseye*); or look for a geometric contradiction in each candidate and eliminate it (*elimination*). Accuracy is 23.8% for *birdseye*, 22.5% for *mental_rotation* and *anchor*, 18.8% for *elimination*, 17.5% for *cot* and 13.8% for *cot + view*. None reaches chance (see Figure 3).

Models fail on the same trials, and their errors ignore layout. If the models were failing for unrelated reasons they would fail on unrelated trials, which they do not. Cohen’s κ on trial-level correctness, over all 100 trials, averages 0.18 across the 120 model pairs and 0.05 across the 16 human–model pairs (permutation $p = 5 \times 10^{-4}$; Appendix Figure 4), so the models resemble each other more than any of them resembles a human. Where the errors land is also informative. Each wrong answer picks one of three foils, and we rank those by layout distance to the target: nearest, middle, farthest. An observer with a graded sense of layout should confuse a place with its nearest neighbour most often. Pooled over 1092 model errors the split is 30/37/33, against the 33/33/33 expected from choices made without regard to geometry, and no individual model departs from a third (Appendix Figure 4), while a human picks the *nearest* foil 10 out of 14 times.

5 Discussion

Every model we tested identifies a place from the studied viewpoint and then loses it once the camera moves. Perception is not the bottleneck as we show that with six landmarks, Qwen2.5-VL-72B identifies 100% of unrotated scenes and answers 19% of the rotated ones, and InternVL3.5-38B behaves the same way. A model that resolves every object in the scene well enough to recognise it from the studied viewpoint has demonstrably seen what it needs to see. Scene complexity is not the

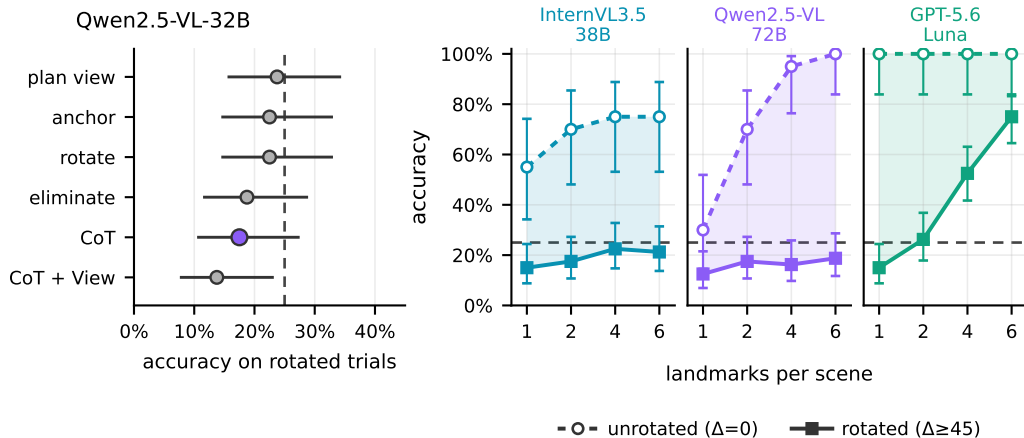


Figure 3: Left: Qwen2.5-VL-32B under six instruction styles, rotated trials only, with the instruction used everywhere else highlighted. The styles are, from the top, "imagine a plan view", "pick an anchor landmark", "rotate the scene mentally", "eliminate alternatives", plain chain of thought, and chain of thought with the viewpoint change asserted. None reaches chance. **Right:** accuracy against the number of landmarks in a scene, one panel per model. The shaded band is the gap between identifying a place and recognising it after a viewpoint change. Adding landmarks raises the unrotated line in all three models and closes the gap in only one. Landmark count and layout separation cannot be varied independently: the median distractor sits at 0.9, 7.6, 23.2 and 28.6 m for one, two, four and six landmarks.

190 bottleneck either: the landmark-count banks vary the scenes instead of the distractors and produce
 191 the same split, with rotated accuracy tracking how far apart the layouts are rather than how much is
 192 in them.

193 The trials are answerable. A human observer, given the same 100 items and the distractors at their
 194 closest spacing, answers 82% of the rotated ones. The information needed is present in the images,
 195 and enough of it survives the viewpoint change for at least one observer to use.

196 Scale does not supply the missing operation. Between 3B and 72B, Qwen2.5-VL gains 45 points on
 197 the appearance gate and loses 9 on rotated trials, and InternVL3.5 repeats the pattern from 1B to 38B
 198 without ever clearing chance. Neither does inference-time computation: the thinking variant of
 199 Qwen3-VL-235B reaches the same 19% rotated accuracy as instruct. We also tested instruction
 200 tuning using six different styles, including two that hand the model an explicit procedure for rotating
 201 the scene or anchoring on a landmark.

202 What is left is the resolution of the representation itself, and the distractor manipulation measures it.
 203 Moving the alternatives from 6.8 m to 31.4 m apart, on the same questions with the same target and
 204 the same answer positions, takes Gemini 3.8 Flash from 39% to 85% and GPT-5.6 Luna from 31% to
 205 55%. A model holding no layout information could not gain 46 points from a change that only moves
 206 the distractors, and a model holding layout at human resolution would not need 30 m of separation
 207 before it could use it. The allocentric map exists but its coarse.

208 **Limitations.** The scaling series are restricted to open-weight models, as frontier parameter counts
 209 remain undisclosed. In the landmark-count banks, the number of landmarks and the layout separation
 210 co-vary, meaning the axis measures layout separation rather than set size in isolation. Finally, as the
 211 stimuli are synthetic, exposure to similar rendered environments during pre-training could provide
 212 advantages. The human baseline is based on one participant which establishes that the task is solvable.
 213 More participants are currently being evaluated.

214 References

215 [1] Chris M. Bird, Dennis Chan, Tom Hartley, Yolande A. Pijenburg, Martin N. Rossor, and
 216 Neil Burgess. Topographical short-term memory differentiates Alzheimer's disease from
 217 frontotemporal lobar degeneration. *Hippocampus*, 20(10):1154–1169, 2010.

- [2] Dennis Chan, Laura M. Gallaher, Kuven Moodley, Ludovico Minati, Neil Burgess, and Tom Hartley. The 4 Mountains Test: a short test of spatial memory with high sensitivity for the diagnosis of pre-dementia Alzheimer’s disease. *Journal of Visualized Experiments*, (116): e54454, 2016.
- [3] Markus Frey, Christian F Doeller, and Caswell Barry. Probing neural representations of scene perception in a hippocampally dependent task using artificial neural networks. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2113–2121. IEEE, 2023.
- [4] Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: multimodal large language models can see but not perceive. In *European Conference on Computer Vision (ECCV)*, 2024.
- [5] Bo Gu, Zhikang Zhang, Zizhuang Wei, Zhenyuan Chen, Lingyun Li, and Zhuoyi Song. Space-Mind++: toward allocentric cognitive maps for spatially grounded video MLLMs. *arXiv preprint arXiv:2605.09449*, 2026.
- [6] Tom Hartley, Chris M. Bird, Dennis Chan, Lisa Cipolotti, Masud Husain, Faraneh Vargha-Khadem, and Neil Burgess. The hippocampus is required for short-term topographical memory in humans. *Hippocampus*, 17(1):34–48, 2007.
- [7] Dingming Li, Hongxing Li, Zixuan Wang, Yuchen Yan, Hang Zhang, Siqi Chen, Guiyang Hou, Shengpei Jiang, Wenqi Zhang, Yongliang Shen, Weiming Lu, and Yueting Zhuang. ViewSpatial-Bench: evaluating multi-perspective spatial localization in vision-language models. *arXiv preprint arXiv:2505.21500*, 2025.
- [8] John O’Keefe and Lynn Nadel. *The Hippocampus as a Cognitive Map*. Oxford University Press, 1978.
- [9] Shouwei Ruan, Bin Wang, Zhenyu Wu, Qihui Zhu, Yuxiang Zhang, Hang Su, and Yubin Wang. World2Mind: cognition toolkit for allocentric spatial reasoning in foundation models. *arXiv preprint arXiv:2603.09774*, 2026.
- [10] Roger N. Shepard and Jacqueline Metzler. Mental rotation of three-dimensional objects. *Science*, 171(3972):701–703, 1971.
- [11] Edward C. Tolman. Cognitive maps in rats and men. *Psychological Review*, 55(4):189–208, 1948.
- [12] Ruth A. Wood, Kuven K. Moodley, Colin Lever, Ludovico Minati, and Dennis Chan. Allocentric spatial memory testing predicts conversion from mild cognitive impairment to dementia: an initial proof-of-concept study. *Frontiers in Neurology*, 7:215, 2016.
- [13] Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in space: how multimodal large language models see, remember, and recall spaces. *arXiv preprint arXiv:2412.14171*, 2024.
- [14] Hantao Zhang, Jinru Sui, Ed Li, Dirk Bergemann, and Zhuoran Yang. MultiView-Bench: a diagnostic benchmark for world-centric multi-view integration in VLMs. *arXiv preprint arXiv:2607.08970*, 2026.
- [15] Yuyou Zhang, Radu Corcodel, Chiori Hori, Anoop Cherian, and Ding Zhao. SpinBench: perspective and rotation as a lens on spatial reasoning in VLMs. In *International Conference on Learning Representations (ICLR)*, 2026.

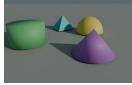
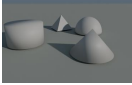
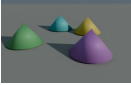
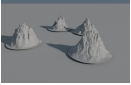
					
Observer	c0	c1	c2	c3	c4
Human	94%	75%	81%	81%	81%
InternVL3.5-4B	19%	6%	25%	0%	12%
InternVL3.5-2B	12%	12%	19%	25%	31%
InternVL3.5-1B	12%	19%	25%	25%	19%
InternVL3.5-38B	19%	6%	19%	19%	31%
InternVL3.5-8B	19%	6%	19%	44%	19%
InternVL3.5-30B-A3B	6%	19%	19%	31%	38%
Qwen2.5-VL-32B	12%	0%	31%	6%	38%
Qwen2.5-VL-3B	38%	12%	31%	38%	25%
Qwen3-VL-235B (t)	12%	12%	19%	19%	31%
InternVL3.5-14B	25%	6%	31%	38%	31%
Qwen2.5-VL-72B	19%	12%	12%	31%	25%
Qwen3-VL-235B (i)	12%	6%	19%	25%	31%
InternVL3-38B	25%	12%	12%	38%	19%
Qwen2.5-VL-7B	50%	19%	25%	19%	44%
Gemini 3.8 Flash	38%	38%	44%	44%	31%
GPT-5.6 Luna	38%	19%	50%	31%	19%
All 16 models pooled	22%	13%	25%	27%	28%

Table 3: Rotated-trial accuracy by stimulus mode, 16 trials per cell, with one place shown in each mode above its column. c0 shape and colour, c1 shape only, c2 colour only, c3 bare peaks, c4 valley. Rows are ordered by overall accuracy. Bold marks the human row and, for each model, its weakest mode; ties go to the rightmost.

A Additional results

Table 3 splits the rotated trials by stimulus mode. Four of the five modes sit within each other’s confidence intervals. The exception is c1, shape without colour, where the models answer 33 of 256 trials correctly (12.9%, Wilson [9.3, 17.6]) against 25.5% [22.9, 28.2] pooled over the other four (two-proportion $z = 4.3$, $p = 2 \times 10^{-5}$). It is the weakest cell for six of the sixteen models outright and joint-weakest for six more. Colour alone (c2) costs nothing by comparison, at 25.0%. It is consistent with colour carrying more identity information than shape in these renders, which would make c1 the mode with the least appearance to work from. It does not bear on the viewpoint result: every mode is far below the human, at every distractor separation.

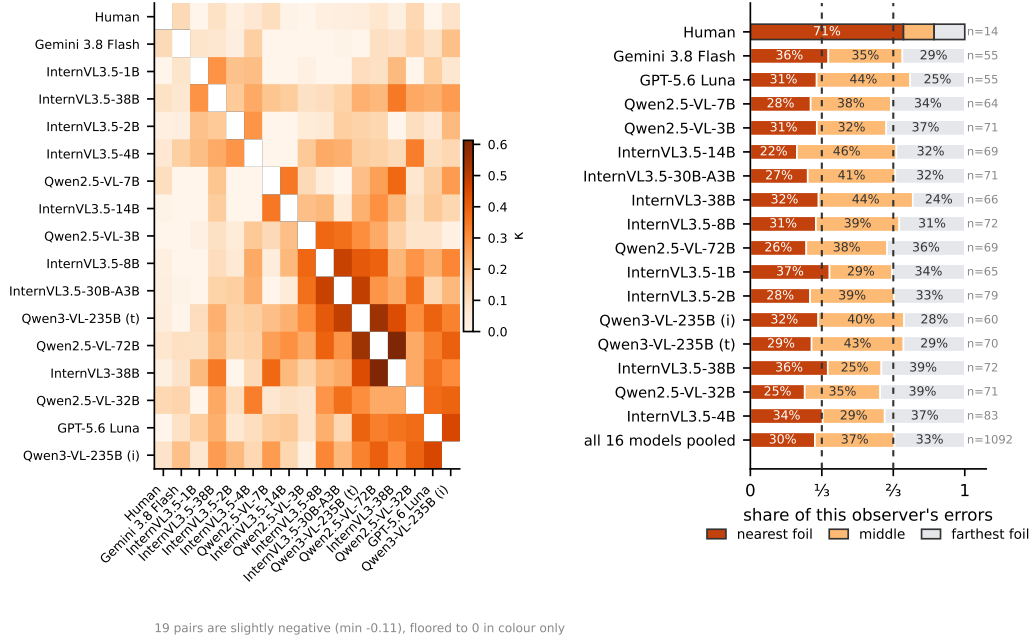


Figure 4: **Left:** do a pair of observers agree on which trials are solvable? Cohen’s κ on trial-level correctness; higher values mean a pair succeeds and fails on the same trials. The human observer is in the first row, with models ordered by mutual similarity. **Right:** when an observer is wrong, which distractor does it pick? Distractors are ranked by layout distance to the target, and the dashed lines mark thirds, which is what errors independent of layout geometry would give.

A.1 The instructions, verbatim

Every model saw one of these, followed by the study image and the four options. The sweep in Figure 3 runs the first six on Qwen2.5-VL-32B; every other run in the paper uses cot_anyview. Lines are wrapped to fit the page.

cot_anyview chain of thought, viewpoint unstated.

You are taking the Four Mountains Test of spatial allocentric perception.

Image 1 is the STUDY view of a landscape with four mountain peaks.

The subsequent 4 images are OPTION 1 to OPTION 4.

Exactly ONE option shows the EXACT SAME mountain landscape (the same four peaks in the same relative spatial arrangement). It may be viewed from the same direction as the study image or from a different one, and the lighting/weather may differ. The other options show different mountain landscapes.

In 2-4 sentences, compare the 3D spatial layout of the peaks (e.g. relative positions such as in front, behind, left, right) between the study view and the options, accounting for any camera rotation. Avoid lengthy itemized lists.

State your final decision on the last line as:

Final Answer: Option X

cot chain of thought, viewpoint asserted.

You are taking the Four Mountains Test of spatial allocentric perception.

Image 1 is the STUDY view of a landscape with four mountain peaks.

The subsequent 4 images are OPTION 1 to OPTION 4.

Exactly ONE option shows the EXACT SAME mountain landscape (the same four peaks in the same relative spatial arrangement), simply viewed from a different viewpoint and under different lighting/weather.

297 The other options show different mountain landscapes.
 298
 299 In 2-4 sentences, compare the 3D spatial layout of the peaks (e.g. relative
 300 positions such as in front, behind, left, right) between the study view and the
 301 options, accounting for camera rotation. Avoid lengthy itemized lists.
 302
 303 State your final decision on the last line as:
 304 Final Answer: Option X

305 mental_rotation instructed to mentally rotate.

306 You are taking the Four Mountains Test of allocentric spatial perception.
 307
 308 Image 1 is the STUDY view of a landscape with four distinct mountain peaks.
 309 The subsequent 4 images are OPTION 1 to OPTION 4.
 310 Exactly ONE option shows the EXACT SAME mountain landscape (the same four peaks in
 311 the same relative geometric layout), viewed from a shifted camera viewpoint and
 312 under different weather/lighting.
 313 The other options are distractors where the relative 3D spatial arrangement of the
 314 peaks has been altered.
 315
 316 To determine the matching scene, perform mental rotation:
 317 1. Estimate the camera perspective shift (rotation angle) between the study image
 318 and candidate options.
 319 2. Mentally rotate the four peaks to verify if topological handedness (e.g.
 320 clockwise/counterclockwise order of peaks, which peak is opposite or between
 321 others) matches the study scene.
 322 3. Ignore superficial differences in sunlight, shadows, fog, and seasonal color.
 323
 324 In 2-4 sentences, explain your mental rotation reasoning, then conclude on the last
 325 line:
 326 Final Answer: Option X

327 anchor instructed to pick an anchor landmark.

328 You are taking the Four Mountains Test of allocentric spatial perception.
 329
 330 Image 1 is the STUDY view of a landscape with four distinct mountain peaks.
 331 The subsequent 4 images are OPTION 1 to OPTION 4.
 332 Exactly ONE option shows the EXACT SAME mountain landscape under a camera viewpoint
 333 rotation and weather change.
 334 The remaining options are distractors.
 335
 336 Spatial Strategy:
 337 1. Select the single most prominent or unique landmark peak as an anchor (origin).
 338 2. Trace the relative bearings and distances of the other 3 peaks surrounding this
 339 anchor.
 340 3. Identify which option preserves this exact 3D spatial configuration around the
 341 anchor under the new camera angle.
 342
 343 In 2-4 concise sentences, explain your reasoning and conclude on the last line:
 344 Final Answer: Option X

345 birdseye instructed to imagine a plan view.

346 You are taking the Four Mountains Test of allocentric spatial perception.
 347
 348 Image 1 is the STUDY view of a landscape with four mountain peaks.
 349 The subsequent 4 images are OPTION 1 to OPTION 4.
 350 Exactly ONE option shows the EXACT SAME mountain landscape viewed from a different
 351 camera viewpoint and lighting.
 352 The other options are distractors with altered 3D mountain configurations.
 353
 354 Top-Down Cognitive Mapping Strategy:
 355 1. Imagine looking down at the four peaks from directly above (a 2D bird's-eye map).
 356 Note their relative positions (which forms a triangle, which is isolated, which is
 357 tallest).

358 2. For each candidate option, determine where the camera would be standing on that
359 same bird's-eye map.
360 3. Verify which option is geometrically consistent with the study scene's top-down
361 layout under the new camera angle.
362
363 In 2-4 sentences, describe the bird's-eye spatial layout and conclude on the last
364 line:
365 Final Answer: Option X

366 elimination instructed to eliminate alternatives.

367 You are taking the Four Mountains Test of allocentric spatial perception.
368
369 Image 1 is the STUDY view of a landscape with four mountain peaks.
370 The subsequent 4 images are OPTION 1 to OPTION 4.
371 Exactly ONE option shows the EXACT SAME mountain landscape viewed from a different
372 camera angle and lighting.
373 The other options are geometric distractors.
374
375 Falsification Strategy:
376 1. Inspect each candidate option one by one to find geometric contradictions with
377 the study scene (e.g. impossible relative peak heights, wrong peak ordering, or
378 missing ridges).
379 2. Eliminate the distractor options that cannot possibly match the study landscape
380 under any viewpoint rotation.
381 3. Select the remaining single candidate that has no geometric contradictions.
382
383 Briefly eliminate the distractors and conclude on the last line:
384 Final Answer: Option X

385 neutral_anyview the human arm, and the only wording a person saw.

386 You will see a STUDY image of a place, then 4 options.
387
388 The place contains several landmarks. Exactly ONE option shows the SAME place as the
389 study image. It may be photographed from the same direction as the study image or
390 from a different one, and the lighting may differ.
391 The other options show different places, each containing the same landmarks arranged
392 differently, photographed from the same direction as the correct option.
393
394 Which option shows the same place as the study image?
395 Answer on the last line as:
396 Final Answer: Option X